IQP Generative Models for Strong Gravitational Lensing Images

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Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed hard to sample classically [4, 5], while the expectation values a maximum-mean-discrepancy (MMD) loss needs stay computable, so such a Born machine trains classically and deploys on quantum hardware [15]. A recent correlation–complexity map predicts which data suit them, and DeepLense strong-lensing images sit at the most IQP-compatible position we measured, above the turbulence data it was introduced on. The reference protocol's fixed latent block yields only noise and a learned linear map loses coherence past $M{=}6$, so we let a small non-linear map carry the latent. At eleven training images the resulting 10-qubit machine matches a DCGAN of six times as many parameters on both generation metrics within the seed spread, leads it on a flux-aligned variant, leads an RBM on all three and reconstructs unseen images at correlation 0.977; at 100 images the DCGAN leads. Every circuit is simulated exactly at $n = 10$ qubits over five seeds, so we claim no quantum advantage, only a step toward utility where training data are scarce.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard transform of a phase function. Exact classical sampling from them would collapse the polynomial hierarchy [4] and the hardness is conjectured to survive approximate sampling [5], though noise or special structure restores classical simulability in some cases [14, 7]. The same structure keeps an MMD loss's expectation values computable, so such a model can be *trained classically and deployed on quantum hardware* [15, 16].

Whether real data sit where that is useful is empirical, and a recent proposal [11] approaches it with two ingredients. (i) A *correlation–complexity map* placing a dataset at I_{QCLI}, the Jensen–Shannon divergence between its Walsh order spectrum and the i.i.d. binomial baseline, and I_{CCI}, how much of its correlation the best pairwise tree misses [19, 17, 6], the high-I_{QCLI}, high-I_{CCI} corner being where IQP circuits should do well. (ii) A latent-adaptation protocol (Algorithm 1 of that paper, a phase core trained on one anchor plus a fixed block of 50 Walsh phases adapted sample by sample), both demonstrated on 2D turbulence [10] at $n = 18$ qubits.

We take both to simulated dark-matter strong-lensing images from DeepLense [1, 3] and contribute three things. We recompute the map on lens crops, which land at its most IQP-compatible position, and show with a permutation null that such a position is necessary and not sufficient. We trace why the reference latent protocol produces noise on static images and replace its latent map by a non-linear one. We then compare against an RBM and a DCGAN under two intensity conventions, and report a replication on the turbulence data in App. D.

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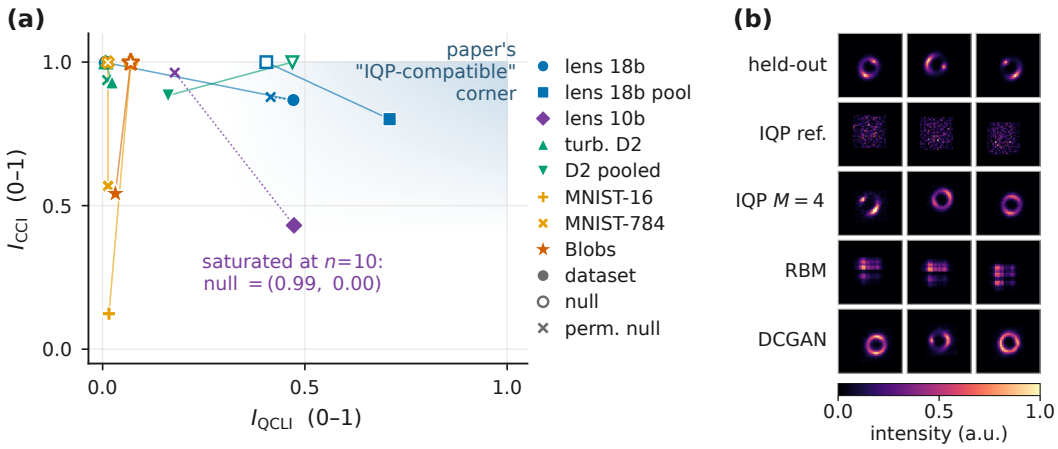


Figure 1: **(a)** Correlation–complexity map, datasets filled, i.i.d.-uniform nulls open, permutation nulls as crosses (ours, not in the paper); lens 18b and 18b pool are the per-image and pooled 18-bit encodings, lens 10b the crop pmf, turb. D2 the paper's turbulence data. **(b)** First three unconditional samples per model, $N{=}100$, no selection.

2 Setup

Model. Every arm is a 10-qubit IQP Born machine over 1,024 Walsh phases θ, differing only in how a per-image latent enters. The reference protocol freezes a random block of 50 of those phases and adapts it sample by sample, which on static lens crops leaves the other training images poorly reconstructed (Sec. 3.2), so we make that map trainable, first as a linear map of M directions,

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M\times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

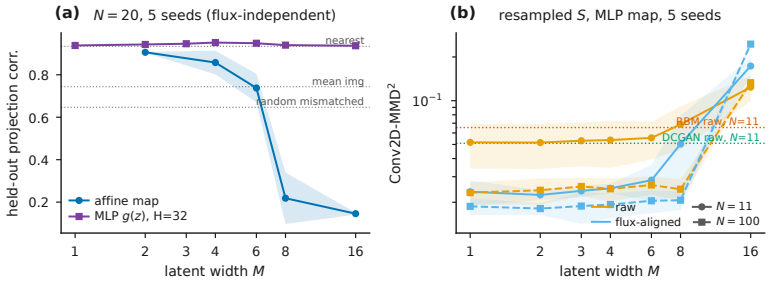
with each row of Φ normalised in L^2. That map degrades once the $\|z\|$ the images demand leaves the radius over which the Walsh phases interfere coherently, by $M{=}6$ on these crops, so our main arm replaces $\Phi^\top z$ by a one-hidden-layer $\tanh$ map g of width $H{=}32$, the z_i trained jointly with g under a small L^2 penalty. The objectives are the flat MMD (uniform Walsh weights, estimable from hardware shots) and the paper kernel (the multi-bandwidth Gaussian MMD kernel taken from IQPopt [16]), and θ^0 is warm-started by 1,000 exact-KL steps, which a flat-MMD warm start replaces at no cost (App. E).

Decoding, baselines and metrics. A trained pmf is a shape, so it is pasted into a 64×64 frame at a total intensity S and a training-set offset. The reference convention gives every frame the training-set mean S, leaving the generated flux with zero variance where real crops spread by about a quarter of their mean, so we draw S per image from a kernel density estimate of the N training totals, with no retraining; App. B repeats everything under the fixed S. The baselines are a restricted Boltzmann machine (RBM) [9, 18] and a deep convolutional generative adversarial network (DCGAN) [13], both early-stopped on 200 validation images while the IQP arms run a fixed 4,000 steps, which can only favour the baselines. We score with two metrics, the Jensen–Shannon divergence between intensity histograms (PDF–JS) and Conv2D-MMD^2 in a fixed random encoder's feature space [8], and since the arms do not agree on how flux is set we add a post-hoc flux-aligned Conv2D-MMD^2 rescaling every frame to the held-out mean total, plus a held-out projection correlation taken over eight restarts at the lowest fit loss. Bins, scale and bandwidth are set on the held-out data alone, so no metric moves with the arm scored, and every arm generates 200 images against the same 200 held-out frames at $N \in \{11, 100\}$, every number a mean $\pm$ sample sd over five seeds above a real-vs-real floor.

Departures from the reference setup. Five choices differ, each following from the data. Its snapshots interpolate to unseen times, whereas the DeepLense `Model II` images we use [1, 2, 12] are static, so we generate unconditionally from the latent prior. It encodes a sample as 18-bit $(x, y, \text{intensity})$ tuples, whereas we centroid-align each image and read its central 32×32 crop, 99.4% of the light, as one probability mass function (pmf) over 1,024 cells at $n = 10$ (App. A), a raw intensity axis carrying barely one bit. It fixes the latent block, whereas we learn the map and

Table 1: Correlation–complexity map, each permutation null keeping the value histogram of the row above, five seeds. Both indices depend on K, the number of n-bit codes per image, so a row reads only against its own-K null; at the paper's own $K = 5 \times 10^4$ the turbulence I_{QCLI} is 0.091 against a null of 0.098.

Dataset and encoding	K	I_{QCLI}	I_{CCI}
Lens, 18-bit $(x, y, \mathrm{intensity})$ tuples	4,096	0.4715 ± 0.0000	0.8677 ± 0.0000
permutation null	4,096	0.4154 ± 0.0000	0.8783 ± 0.0001
Turbulence D2, 18-bit tuples	4,096	0.0232 ± 0.0002	0.9285 ± 0.0008
permutation null	4,096	0.0114 ± 0.0000	0.9371 ± 0.0009
Lens, 10-bit crop pmf	exact pmf	0.4731 ± 0.0000	0.4313 ± 0.0000
permutation null	exact pmf	0.1782 ± 0.0000	0.9634 ± 0.0004



(a) The two curves come from different drivers, the affine one from the mechanism test and the MLP one from the round-3 protocol, on the same 20 training and 20 held-out frames, with shared reference lines.
(b) One driver throughout; colour encodes the metric, not the map, and the aligned curve is identical under either flux convention by construction.

Figure 2: Latent width. **(a)** Held-out projection correlation against M on a shared 20-image split, five seeds, with that split's reference levels; the linear curve comes from the mechanism-test driver and the non-linear one from the round-3 trainer, both scored on the same 20 held-out frames. **(b)** Raw and flux-aligned Conv2D-MMD^2 under the sampled intensity, with the baseline levels of Table 2 at N=11.

train the N images jointly, and it estimates its MMD from samples where a 10-qubit pmf is exact. It holds out times, so we project onto held-out images instead.

3 Results

3.1 Lens images are the most IQP-compatible dataset on the map

Table 1 recomputes the map. Encoded as the reference paper encodes turbulence, lens images sit far above it in I_{QCLI} and inside the corner the map calls IQP-compatible, yet a permutation null keeping each image's intensity histogram nearly matches them, and turbulence behaves the same way, so at this encoding the map largely tracks the histogram. Only the 10-bit crop pmf separates from its null in both coordinates, so a favourable position is necessary, not enough.

3.2 From the reference protocol to a non-linear latent map

Run sequentially on our low-variance latent block, Algorithm 1 produces noise (Fig. 1(b), second row). Representability is not the cause, since the full 1,024-phase circuit fits a *single* aligned crop to correlation 0.99997 ± 0.00001. The protocol is, since with the core frozen on one anchor the 50 fixed coordinates reconstruct every other training image at only 0.44 ± 0.04, fixed Walsh directions cross-fading pixels as an XOR-shift pattern instead of moving lens structure. Equation (1) repairs that but introduces a width limit of its own. On a 20-image sweep the learned linear map holds a held-out projection correlation of 0.9060 ± 0.0021 at M=2, falls to 0.7378 ± 0.0672 at M=6 and collapses to 0.2184 ± 0.1213 at M=8. The non-linear map removes that collapse, holding 0.9368 to 0.9519 across the whole sweep, above the nearest-training-image reference of 0.9339 (Fig. 2(a)), but not the small useful width, since on the generation metrics it is flat to M=4 and degrades from M=8. We report it at M=2, where the held-out projection correlation is 0.9013 ± 0.0012 at N=11 and 0.9773 ± 0.0006 at N=100.

Table 2: Comparison with the classical baselines on lens crops under the sampled intensity, five seeds, PDF–JS in 10^{-3} and both MMD^2 columns in 10^{-2}. Lower is better; **bold** marks a best mean leading every other arm by more than both sds, so a column without bold carries no win the seed spread supports. Params are totals at N=11 / N=100, the last row a real-vs-real floor, and App. E gives † and the parameter convention.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD$^2_{\text{al}}$	PDF–JS	MMD^2	MMD$^2_{\text{al}}$
IQP, reference protocol (Alg. 1)†	1,524 / 5,974	6.6 ± 0.8	14.8 ± 1.6	22.5 ± 2.2	7.7 ± 0.4	14.7 ± 0.3	26.6 ± 0.7
IQP, fixed basis, joint	1,524 / 5,974	1.8 ± 0.4	8.7 ± 1.4	12.5 ± 1.3	1.1 ± 0.2	5.4 ± 0.4	8.5 ± 1.5
IQP, linear map M=4 (flat MMD)	5,164 / 5,520	2.8 ± 0.1	5.6 ± 1.9	2.7 ± 0.2	2.6 ± 0.2	2.8 ± 0.5	2.6 ± 0.4
IQP, linear map M=4 (paper kernel)	5,164 / 5,520	1.0 ± 0.05	6.3 ± 1.5	7.4 ± 0.6	1.5 ± 0.2	5.9 ± 0.4	10.0 ± 0.8
IQP, non-linear map M=2 (flat MMD)	33,910 / 34,088	1.2 ± 0.07	5.2 ± 1.8	**2.2** ± 0.2	1.7 ± 0.2	2.4 ± 0.5	1.8 ± 0.2
IQP, non-linear map M=2 (paper kernel)	33,910 / 34,088	0.75 ± 0.07	5.4 ± 1.5	5.1 ± 0.3	0.99 ± 0.04	4.2 ± 0.4	6.6 ± 0.6
RBM [18]	11,510 / 15,960	1.9 ± 1.0	7.9 ± 1.9	11.1 ± 3.9	1.1 ± 0.2	5.1 ± 0.8	8.1 ± 2.2
DCGAN [13]	199,837	1.1 ± 0.6	5.1 ± 1.5	3.9 ± 1.0	**0.55** ± 0.12	2.2 ± 0.3	2.1 ± 0.4
Real-vs-real floor	n/a	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2

3.3 Comparison with classical baselines

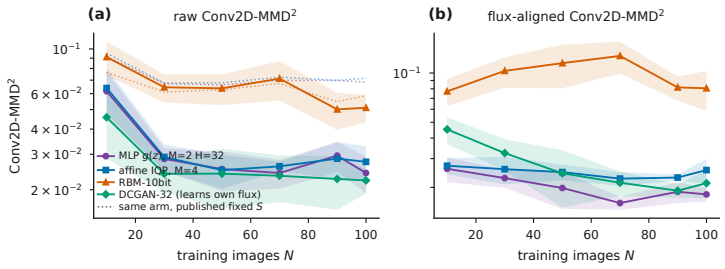
In Table 2 the eleven-image regime is where the IQP arms are competitive. There the non-linear arm matches the DCGAN on PDF–JS and raw Conv2D-MMD^2 within seed noise (-0.13 and -0.04 of the DCGAN's own sd) and leads on the flux-aligned column ($+1.57$ sd), and it leads the RBM on all three by 0.75, 1.40 and 2.27 of the RBM's sd, with six times fewer parameters. Under the paper kernel the same map holds the table's best mean PDF–JS, 0.75×10^{-3} against the DCGAN's 1.11 ± 0.56, which leaves the two inside one sd, and trades that for a worse flux-aligned column. No arm beats the DCGAN on all three columns by more than a seed sd at either size, and at 100 images the DCGAN leads on PDF–JS and raw Conv2D-MMD^2, the flux-aligned 1.8 ± 0.2 sitting inside one sd of its 2.1 ± 0.4.

Two properties bound how the table reads. The flux-aligned column is invariant to the intensity convention rather than independent of the raw one, the largest paired difference there being 6×10^{-9} over our 325 runs, so the two MMD^2 columns measure shape, and shape plus flux, not two separate things. The sampling rule is also not neutral between arms, improving every IQP arm's raw Conv2D-MMD^2 by 22 to 39% at N=11 while making the RBM's 20% worse and improving it by 13% at N=100, so part of the eleven-image gap comes from the baseline moving. Sampling fixes the shape of the generated flux distribution and not its location, the eleven training frames being 9% dimmer than the held-out set, which is why the raw column stops beside the DCGAN.

Data scaling and per-class training. Across $N \in \{10, \ldots, 100\}$ (Fig. 3) the non-linear arm's raw Conv2D-MMD^2 falls from 0.062 ± 0.016 to 0.024 ± 0.005 and the RBM's from 0.091 ± 0.017 to 0.051 ± 0.008, the DCGAN starting ahead at N=10 (0.046 ± 0.017) and ending level. On the flux-aligned column the non-linear arm holds the lowest mean at every N, though only at N=10 and N=70 by more than a seed sd. Retrained inside one class at N=100 it reaches flux-aligned 0.017, 0.015 and 0.021 for axion, cdm and no-sub against floors of 0.0031, 0.0032 and 0.0045 (App. C), so each class can be generated on its own, not that the model separates them.

4 Discussion and limitations

At eleven training images a 10-qubit IQP Born machine with a learned non-linear latent map is competitive with an RBM and a DCGAN and on the flux-aligned metric surpasses both at a sixth of the DCGAN's parameters, a step toward utility where training data is the scarce resource. Nothing here is evidence of quantum advantage, since every circuit is simulated exactly at $n = 10$ qubits and the hardness results describe the class at scale, and the standing does not survive to 100 images. Five seeds are few, the DCGAN adds a spread of up to 57% on raw Conv2D-MMD^2 at one seed, and the sampled intensity moves the raw column unequally across arms.



Mean ± 1 sd over 5 seeds; MLP $g(z)$ n=5, affine IQP n=5, RBM-10bit n=5 at N=100. Solid = resampled S, dotted = the published fixed S, in panel (a) alone, since the aligned metric in (b) is identical under either convention. DCGAN has one curve because it learns a per-image flux and has no fixed S to swap; the same 200 held-out frames and the same v11 metrics are used throughout.

Figure 3: Conv2D-MMD^2 against training-set size, raw in **(a)** and flux-aligned in **(b)**, five seeds; solid is the sampled intensity, dotted the fixed one.

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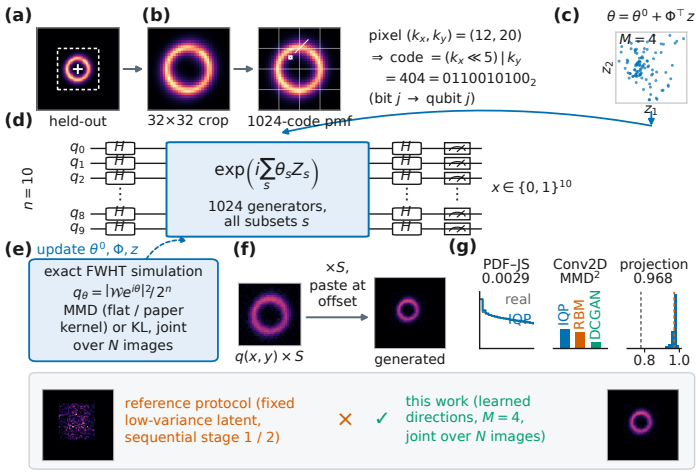


Figure 4: Pipeline. **(a)** A held-out image and the crop window. **(b)** The aligned crop as a pmf over $1{,}024$ cells, the pixel $(12, 20)$ carrying code 404, bit j addressing qubit j. **(c)** Learned latents at $M = 4$. **(d)** The circuit. **(e)** Exact simulation, drawn for Eq. (1); the main arm holds $\theta = \theta^0 + g(z)$ there. **(f)** Decoding at S. **(g)** Metrics for the linear $M{=}4$ arm.

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A Pipeline

B The fixed-S convention

Table 3 is Table 2 with the single training-mean intensity every pmf-based arm published under, on the same runs; rebuilding those frames from each run's saved crops reproduces the published ones exactly, so the two tables differ only in the decode, and the comparison is paired. On raw Conv2D-MMD^2 at $N{=}11$ sampling improves every IQP arm by 0.033 to 0.041 with a paired sd of 0.012 to 0.020, the two omitted fixed-block arms included, while making the RBM worse by 0.013 ± 0.017; at $N{=}100$ it improves every arm including the RBM (0.008 ± 0.008). On PDF–JS the paired effect is at most 7×10^{-4} and mixed in sign, and on the flux-aligned column below 10^{-8}.

C Latent sweeps and per-class results

Hidden width matters less than latent width, $H \in \{4, 8, 16, 32\}$ giving flux-aligned 0.029, 0.022, 0.020 and 0.019 at $M{=}4$, $N{=}100$. Each class is scored against its own floor, $0.03/0.36/0.31$ for axion, $0.03/0.30/0.32$ for cdm and $0.04/0.45/0.45$ for no sub in the table's units, so a column reads down a class.

Table 3: Table 2 under the fixed intensity, same runs, seeds, units, bold rule and floor, the two fixed-block arms left to the text. The DCGAN has no fixed S to swap, so its row is repeated.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD^2_{al}	PDF–JS	MMD^2	MMD^2_{al}
IQP, linear map M=4 (flat MMD)	5,164 / 5,520	3.0 ± 0.2	8.9 ± 0.2	2.7 ± 0.2	2.9 ± 0.3	7.2 ± 0.3	2.6 ± 0.4
IQP, linear map M=4 (paper kernel)	5,164 / 5,520	0.81 ± 0.05	9.6 ± 0.3	7.4 ± 0.6	1.4 ± 0.2	11.3 ± 0.5	10.0 ± 0.8
IQP, non-linear map M=2 (flat MMD)	33,910 / 34,088	0.96 ± 0.09	8.5 ± 0.2	**2.2** ± 0.2	1.9 ± 0.2	6.8 ± 0.1	1.8 ± 0.2
IQP, non-linear map M=2 (paper kernel)	33,910 / 34,088	0.49 ± 0.07	8.7 ± 0.2	5.1 ± 0.3	0.86 ± 0.08	9.8 ± 0.3	6.6 ± 0.6
RBM [18]	11,510 / 15,960	1.7 ± 0.8	6.5 ± 0.8	11.1 ± 3.9	0.92 ± 0.13	5.9 ± 0.4	8.1 ± 2.2
DCGAN [13]	199,837	1.1 ± 0.6	5.1 ± 1.5	3.9 ± 1.0	**0.55** ± 0.12	**2.2** ± 0.3	2.1 ± 0.4

Table 4: Latent-width sweep at H=32 under the flat MMD, then the same arm retrained inside one class; sampled intensity, five seeds, units of Table 2.

	$N = 11$			$N = 100$		
M	PDF–JS	MMD^2	MMD^2_{al}	PDF–JS	MMD^2	MMD^2_{al}
2	1.2 ± 0.07	5.2 ± 1.8	2.2 ± 0.2	1.7 ± 0.19	2.4 ± 0.5	1.8 ± 0.2
4	1.2 ± 0.04	5.4 ± 1.9	2.5 ± 0.2	1.6 ± 0.10	2.5 ± 0.5	1.9 ± 0.4
16	7.7 ± 0.91	12.4 ± 2.5	17.3 ± 2.5	8.8 ± 0.14	13.3 ± 0.3	24.5 ± 0.8
axion	1.1 ± 0.10	4.6 ± 0.6	3.0 ± 0.4	1.8 ± 0.17	2.9 ± 0.2	1.7 ± 0.2
cdm	0.94 ± 0.05	4.4 ± 0.2	2.7 ± 0.3	1.7 ± 0.06	2.1 ± 0.3	1.5 ± 0.3
no sub	0.98 ± 0.10	3.1 ± 0.9	2.4 ± 0.6	1.7 ± 0.10	2.3 ± 0.4	2.1 ± 0.3

D Replication of the reference paper's turbulence experiment

On the reference paper's own dataset (2D cylinder-wake turbulence D2 [10], $n = 18$ qubits, 11 anchor snapshots, five seeds) the result is **not reproduced** under our pre-registered convention, the RBM (PDF–JS 0.217 ± 0.007) and a 1.2M-parameter DCGAN (0.045 ± 0.062) beating our 31k-parameter IQP (0.272 ± 0.004) on *both* metrics. Under a last-write-wins aggregation convention, never specified in the paper, our PDF–JS values land close to the published ones (0.0324 and 0.0473 against 0.0238 and 0.0221), but the RBM still wins; and Conv2D-MMD^2 is saturated, ground truth shuffled in pixel space already scoring 0.7045 ± 0.0095, below every model in that paper's Table 2.

E Training and evaluation details

Table conventions. The † arm runs Algorithm 1 on our low-variance latent block, whose stage 1 trains the phase core and the anchor's latent block jointly where the reference protocol freezes a random one. Params count the shared core plus the per-image latent block at each N, the DCGAN entry is its generator alone, and the RBM holds $10{,}960$ shared weights plus 50 per image; it ran 160k or 80k steps and the DCGAN 5k to 30k.

Protocol details. The paper kernel is the three-bandwidth Gaussian MMD of IQPopt [16] at $\sigma = \{0.989, 1.729, 2.236\}$, that paper publishing none. The Conv2D-MMD^2 bandwidth, fixed once on the held-out features and shared by every table, is 0.005547 raw and 0.002558 flux-aligned, the 20 floor draws come from the $8{,}710$ validation images the held-out split leaves, and the Silverman bandwidth of the intensity estimate is 14.64 at N=11 and 10.36 at N=100. Applying the baselines' validation rule to the IQP arms selects 1k to 16k steps and changes no cell of Table 2 by more than one sd, so we keep the fixed $4{,}000$ steps; at $20{,}000$ fixed steps both baselines are much worse, so their early stopping is load-bearing and is granted throughout. The DCGAN is the only arm here that is not bit-reproducible.

Ablations. Three settings of the main arm were varied and none moves a metric by more than a seed sd. A flat-MMD warm start of θ^0 in place of the exact-KL one, making every stage shot-estimable, leaves flux-aligned Conv2D-MMD^2 at 0.0231 against 0.0225 at N=11; picking the projection restart by correlation rather than by fit loss moves the correlation by 0.0003; and two restarts match eight, though the zero-initialised restart alone reaches only 0.838 and 0.887 against 0.901 and 0.977.